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Math 215A

Homework 4
Due: October 21, 2026

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Let I be an ideal in a commutative ring R, $I \neq R$. Show that there is a minimal prime ideal containing I. (That means: there is a prime ideal containing I which contains no other prime ideal containing I.) Can you translate this statement into geometric terms, about $\mathrm{Spec}(R)$? (Remember that there are two different ways to think of prime ideals in terms of $\mathrm{Spec}(R)$: they are the points of $\mathrm{Spec}(R)$, but they also correspond to the irreducible closed subsets of $\mathrm{Spec}(R)$, in a way you can recall.)

Solution. We will prove this in two steps. We will first show that $(\mathrm{Spec}(R), \subseteq)$ has a minimal element, a slightly more general statement. Then we will show why that implies that set of the prime ideals which contain $I \subseteq R$, an ideal, or $V(I)$ has a minimal element.

1. **$(\mathrm{Spec}(R), \subseteq)$** has a minimal element for any commutative ring $R \neq 0$: Since $R \neq 0$, it has a maximal ideal meaning $\mathrm{Spec}(R) \neq \emptyset$. Let S be a arbitrary chain. Let

$$I = \bigcap_{\mathfrak{p} \in S} \mathfrak{p}.$$

   Since I is the intersection of ideals, we know I is a ideal. Furthermore, we want to show that its prime. Note that $1 \notin \mathfrak{p} \in S \Rightarrow 1 \notin I$. Now suppose $a, b \notin I$, we will show that $ab \notin I$. Since $a, b \notin I$, there must exists a $\mathfrak{p}_a \in S$ such that $a \notin \mathfrak{p}_a$ and $\mathfrak{p}_b \in S$ such that $b \notin \mathfrak{p}_b$. We have two cases:
   1. Suppose $\mathfrak{p}_a \subseteq \mathfrak{p}_b$ (using chain property), $b \notin \mathfrak{p}_a$. Thus $a, b \notin \mathfrak{p}_a \Rightarrow ab \notin \mathfrak{p}_a \Rightarrow ab \notin I$.
   2. Suppose $\mathfrak{p}_b \subseteq \mathfrak{p}_a$, $a \notin \mathfrak{p}_b$. Thus $a, b \notin \mathfrak{p}_b \Rightarrow ab \notin \mathfrak{p}_b \Rightarrow ab \notin I$.

   Thus in either case, we get an implication that $ab \notin I$, meaning that $ab \notin I$. Hence I is prime and $I \subseteq \mathfrak{p}$ for all $\mathfrak{p} \in S$. Hence I is a lower bound for the chain S and $I \in \mathrm{Spec}(R)$. By Zorn's Lemma, since every chain has a lower bound, $\mathrm{Spec}(R)$ has at least 1 minimal element, ie there is a prime ideal $\mathfrak{p}$ of R such that it doesn't contain any other prime ideals.
2. $V(I)$ has a minimal element for $I \subset R$ an proper ideal: Note that

$$V(I) \leftrightarrow \mathrm{Spec}(R/I)$$

   where it sends a prime ideal of R/I to its preimage under the quotient map, which is also a prime ideal. Furthermore, the correspondence respects inclusion. Hence a minimal element of $\mathrm{Spec}(R/I)$ corresponds to a minimal element of $V(I)$, or a prime ideal which contains I but doesn't contain any other prime ideals containing I.

Translating the statement into a topological one, it is equivalent to say that we can partition $\mathrm{Spec}(R)$ into maximal irreducible components.

Problem 2

Let k be a field. Show that every ideal in the polynomial ring $k[x]$ is generated by one element. But show that the ideal $(x^n, x^{n-1}y, ..., y^n)$ in $k[x, y]$ cannot be generated by fewer than $n + 1$ elements. Thus there is no upper bound for the number of elements needed to generate an ideal in $k[x, y]$. (Hint: try to reduce this to a question about the dimension of a vector space over a field.)

Solution. Let I be a ideal in $k[x]$. Note that 0 is a lower bound for $\deg(I)$ and $\deg(I) \subseteq \mathbb{N}$ meaning there exists $f \in I$ such that $\deg(f) \leq \deg(g)$ for all $g \in \frac{I}{\{0\}}$. I will show that $I = k[x]f$. Since $k[x]$ is a Euclidean domain, $\forall g \in I$: $g(x) = p(x)f(x) + q(x)$ for $p, q \in k[x]$ and $\deg(q) < \deg(f)$. $q(x) = g(x) - p(x)f(x) \in I$ since $\deg(f) \leq \deg(q)$ if $q \neq 0$, that means $\deg(f) > \deg(q) \Rightarrow q = 0$. Hence everything in I, is a polynomial multiple of f. Hence $I = k[x]f = (f)$.

Let $\mathfrak{m} = (x, y)$ and let $I_n = (x^n, x^{n-1}y, ..., xy^{n-1}, y^n)$. Let P_n be the k-vector space of degree n homogenuous polynomials in two variables, and note that P_n has dimension $n + 1$. Note that $\mathfrak{m}I_n \subseteq I_n \subset \mathfrak{m}$.. Note the following

$$\{Qh : Q \in k[x, y] - k, h \in P_n\} \text{ generate } \mathfrak{m}I_n.$$

Thus

$$f = 0 \in \frac{I_n}{mI_n} \Leftrightarrow f = \sum_{k=1}^{m} Q_k h_k$$

where $Q_k \in k[x, y] - k$, and $h_k \in P_n$. Thus the only nonzero terms are degree n homogenuous polynomials in 2 variables. In fact all nonzero elements $f \in P_n$ are nonzero in $I_n/\mathfrak{m}I_n$. In less complicated words,

$$I_n/\mathfrak{m}I_n \underset{k \text{ Vect}}{\cong} P_n.$$

Thus if I_n could be generated by less than $n + 1$ elements, so could $I_n/\mathfrak{m}I_n$, or equivalently P_n, which contradicts the fact that the dimension of P_n is $n + 1$, thus a spanning set must have at least $n + 1$ elements.